

7.7.15

Replace section with

External custom-designed girth flanges shall be provided with 3 mm ($1/8$ in.) future machining allowance on the gasket contact seating surface (including pass-partition surfaces). The additional thickness shall not be used in the calculation of maximum allowable working pressure.

This requirement does not apply to clad or weld overlay construction.

Add new section

7.7.17

Maximum spacing between bolt centres shall be as calculated by TEMA RCB-11.22.

7.8 Girth Flange Joint Supplementary Design Requirements

7.8.1

Replace first sentence with

The joint component approach, as defined in ASME PCC-1, Appendix O, and the requirements contained within this section, shall be applied to both sides of the heat exchanger for girth flange and floating head flange joints.

Add new section

7.8.10

For floating head covers, in Step 6 of ASME PCC-1, the following additional check, as calculated by Equation (3), shall be undertaken to verify that the selected assembly bolt stress include the effects of pressure acting from the opposing side of the floating head cover.

$$Sb_{sel} \geq \frac{Sg_{min-o} \times A_{g,FH} + \left[\left[\frac{\pi}{4} \times (G_{ID,FH})^2 \times P_T \right] - \left[\frac{\pi}{4} \times (FH_{OD})^2 \times P_S \right] \right]}{\phi_g \times n_b \times A_b} \quad (3)$$

where

A_b	is the bolt root area, expressed in mm ² (in. ²);
$A_{g,FH}$	is the floating head gasket area, expressed in mm ² (in. ²);
FH_{OD}	is the outside diameter of the floating head flange, expressed in mm (in.);
$G_{ID,FH}$	is the floating head gasket inner diameter, expressed in mm (in.).

7.9 Expansion Joints

7.9.1

Replace first sentence with

Expansion joints in shells shall be thick walled flanged and flued unless otherwise approved by the purchaser.